

ME436A-IS: Plasma Engineering

Joshua Davidov

Paschen Curve Graphing Assignment

Instructor: Professor Wright

April 24, 2026

1. Introduction

In this assignment, we:

- a) Derive $\min(V_{BD})$.
- b) Plot the Paschen curves for air, argon, and carbon dioxide given literature Paschen constants.
- c) Plot points representing the breakdown voltage of air at different selected pd values.
- d) Plot the Paschen curve for air given different secondary Townsend coefficients.

2. Derivation of $\min(V_{BD})$

We claim:

$$\min(V_{BD}) = \frac{Be \ln(\gamma^{-1} + 1)}{A}$$

Recall the equation for Paschen's Law:

$$V_{BD}(pd) = \frac{Bpd}{\ln(Apd) - \ln(\ln(\gamma^{-1} + 1))} \quad (1)$$

To find $\min(V_{BD})$, take the derivative of V_{BD} with respect to pd :

$$\begin{aligned} \frac{dV_{BD}}{d(pd)} &= \frac{B(\ln(Apd) - \ln(\ln(\gamma^{-1} + 1))) - Bpd(pd)^{-1}}{(\ln(Apd) - \ln(\ln(\gamma^{-1} + 1)))^2} \\ \frac{dV_{BD}}{d(pd)} &= \frac{B(\ln(Apd) - \ln(\ln(\gamma^{-1} + 1)) - 1)}{(\ln(Apd) - \ln(\ln(\gamma^{-1} + 1)))^2} \end{aligned} \quad (2)$$

Recall that V_{BD} has a local extrema at $(pd = x)$ if:

$$\left. \frac{dV_{BD}}{d(pd)} \right|_{pd=x} = 0$$

Note that

$$\begin{aligned} &(\ln(Apd) - \ln(\ln(\gamma^{-1} + 1)) - 1 \neq \ln(Apd) - \ln(\ln(\gamma^{-1} + 1))) \\ \Rightarrow &((\ln(Apd) - \ln(\ln(\gamma^{-1} + 1)) - 1 = 0) \Rightarrow (\ln(Apd) - \ln(\ln(\gamma^{-1} + 1)) \neq 0)) \end{aligned}$$

Therefore, we can compute the zero(es) of the derivative of V_{BD} with respect to pd by calculating the zero(es) of $(\ln(Apd) - \ln(\ln(\gamma^{-1} + 1)) - 1)$. Let:

$$0 = \ln(Apd) - \ln(\ln(\gamma^{-1} + 1)) - 1$$

$$\Rightarrow 1 = \ln\left(\frac{Apd}{\ln(\gamma^{-1} + 1)}\right)$$

$$\Rightarrow e = \frac{Apd}{\ln(\gamma^{-1} + 1)}$$

$\Rightarrow pd = \frac{e \ln(\gamma^{-1} + 1)}{A} \quad (3)$

Therefore, $\left(x = \frac{e \ln(\gamma^{-1} + 1)}{A}\right)$ is a local extrema of V_{BD} .

We also claim that $(V_{BD}(x) = \min(V_{BD}))$, which we can prove rigorously by analyzing the second derivative of V_{BD} with respect to pd .

Now, consider the second derivative of V_{BD} with respect to pd (done with Wolfram Alpha):

$$\frac{d^2 V_{BD}}{d(pd)^2} = \frac{B}{pd(\ln(Apd) - \ln(\ln(\gamma^{-1} + 1)))^2}$$

In the context of a Paschen Curve:

$$((p > 0) \wedge (d > 0)) \Rightarrow (pd > 0)$$

Additionally, per Table 1, $(B > 0)$. Since $\left(\left(pd(\ln(Apd) - \ln(\ln(\gamma^{-1} + 1)))^2\right) > 0\right)$, this then implies that $\left(\frac{d^2 V_{BD}}{d(pd)^2} > 0\right)$. Therefore, since the graph of V_{BD} is concave up, then the extrema $\left(x = \frac{e \ln(\gamma^{-1} + 1)}{A}\right)$ is an absolute minimum.

Now, compute V_{BD} at the local extrema present in Equation 3:

$$V_{BD}(x) = \frac{B\left(\frac{e \ln(\gamma^{-1} + 1)}{A}\right)}{\ln\left(A\left(\frac{e \ln(\gamma^{-1} + 1)}{A}\right)\right) - \ln(\ln(\gamma^{-1} + 1))}$$

$$V_{BD}(x) = \frac{B(e \ln(\gamma^{-1} + 1))}{A(\ln(e \ln(\gamma^{-1} + 1)) - \ln(\ln(\gamma^{-1} + 1)))}$$

$$V_{BD}(x) = \frac{B(e \ln(\gamma^{-1} + 1))}{A(1 + \ln(\ln(\gamma^{-1} + 1)) - \ln(\ln(\gamma^{-1} + 1)))}$$

$\min(V_{BD}) = V_{BD}(x) = \frac{B(e \ln(\gamma^{-1} + 1))}{A} \tag{4}$
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3. Graphs/Tables

To produce these graphs, we first have to assume:

- The electric fields between the electrodes are uniform.
- Secondary electrons are produced at the cathode only due to the impact of positive ions created during a Townsend discharge.
- The density of ionized particles is negligible so that it does not affect the uniform electric field between the cathodes.
- The gas between the plates is ideal.

All relevant Paschen Law constants (A , B , and γ) to plot the Paschen curves of relevant gases were derived from Table 4.1 of *Plasma Chemistry*¹ by Alexander Fridman.

Gas	A (torr ⁻¹ · cm ⁻¹))	B (V · torr ⁻¹ · cm ⁻¹)
He	3	34
Ar	12	180
Air	15	365
Ne	4	100
CO ₂	20	466

Table 1: *Table of Paschen Constants, derived from Plasma Chemistry by Alexander Fridman*

The selected air Paschen curve points are:

$$pd \in \{(760.00 \text{ mbar} \cdot 2 \text{ mm}), (966.86 \text{ mbar} \cdot 2 \text{ mm})\}.$$

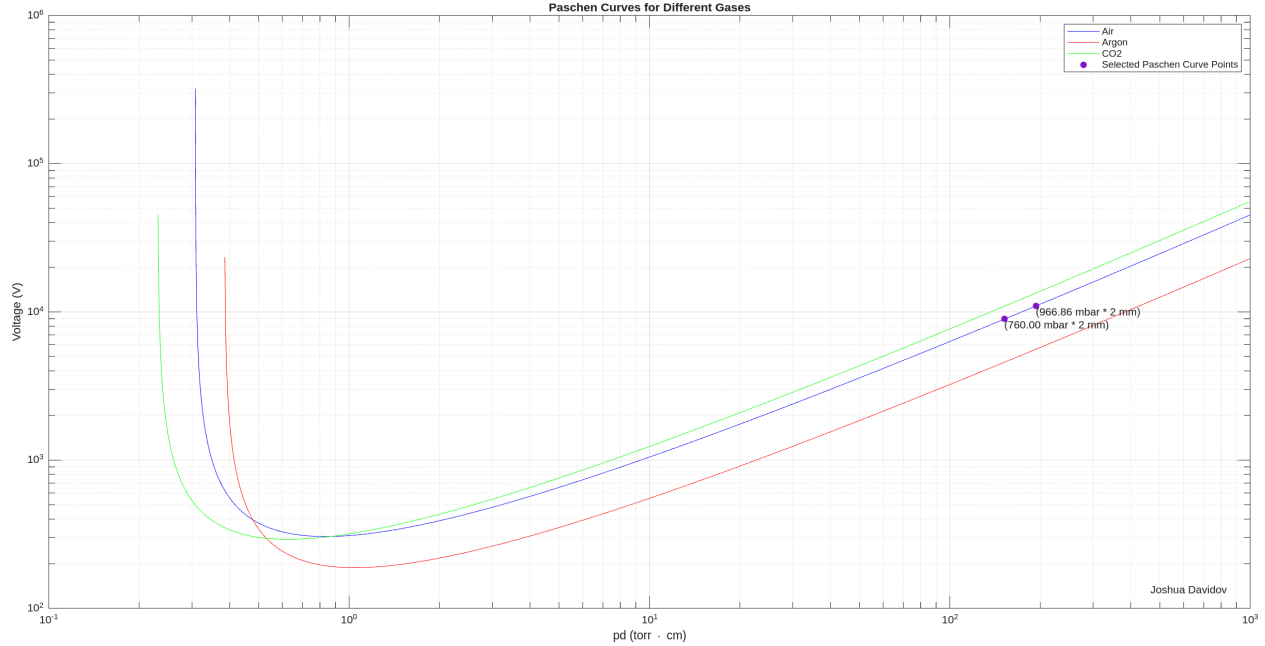


Figure 1: *Paschen Curves for Air, Argon, and Carbon Dioxide, with Selected Air Breakdown-Voltage Points*

As per [“A Townsend’s secondary ionization coefficient estimation method for partial discharge inception voltage prediction for insulating polymers”](#)², there exists materials that, if they were to be used as electrodes in a plasma system similar to the one used in our Plasma lab, would have a secondary Townsend coefficient of ($\gamma \in [10^{-5}, 10^{-1}]$). Therefore, consider the following figure containing the Paschen curve of air for secondary Townsend coefficients ($\gamma \in \{10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$).

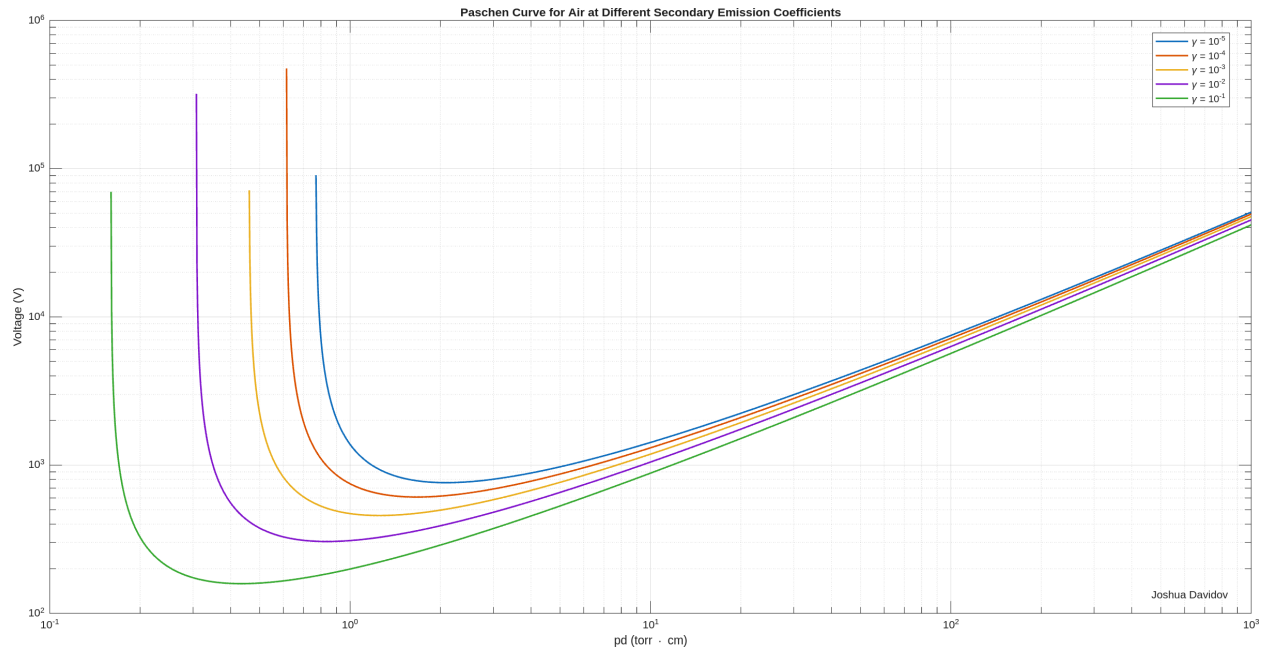


Figure 2: *Paschen Curve of Air of Selected Secondary Townsend Coefficients*

Note: This behavior makes mathematical sense, as $\left(V \propto \log\left(\log\left(\frac{1}{\gamma}\right) + 1\right)\right)$, and physical sense, as a higher secondary Townsend coefficient results in a greater average amount of secondary electrons released per successful atom collision, which increases the likelihood of Townsend discharges to occur.

4. Appendix MATLAB Code

Here was the following code to generate the MATLAB figure:

```
%% Params. for Paschen Curve
pd = logspace(-1,3,2000); % 0.1 to 1000 on a log-spaced grid
gamma = 0.01; %dimensionless; assuming electrodes are made out of stainless
steel
mbartotorr = 0.750062;
psitotorr = 51.7149;
Patm = 1013.25*mbartotorr %torr
x = [Patm*0.2, (Patm+(4*psitotorr))*0.2] %torr*cm
labels = {"(760.00 mbar * 2 mm)", "(966.86 mbar * 2 mm)"};

% Air
A_Air = 15; %m*bar^(-1)*cm^(-1)
B_Air = 365; %V*bar^(-1)*cm^(-1)
V_Air = B_Air .* pd ./ (log(pd) + log(A_Air) - log(log((1 / gamma) + 1)));
V_Air((log(pd) + log(A_Air) - log(log((1 / gamma) + 1))) <= 0) = NaN;

% Ar
A_Ar = 12; %m*bar^(-1)*cm^(-1)
B_Ar = 180; %V*bar^(-1)*cm^(-1)
V_Ar = B_Ar .* pd ./ (log(pd) + log(A_Ar) - log(log((1 / gamma) + 1)));
V_Ar((log(pd) + log(A_Ar) - log(log((1 / gamma) + 1))) <= 0) = NaN;

% CO2
A_CO2 = 20 %m*bar^(-1)*cm^(-1)
B_CO2 = 466 %V*bar^(-1)*cm^(-1)
V_CO2 = B_CO2 .* pd ./ (log(pd) + log(A_CO2) - log(log((1 / gamma) + 1)));
V_CO2((log(pd) + log(A_CO2) - log(log((1 / gamma) + 1))) <= 0) = NaN;

% Plot the Paschen curves for Air, Argon, and CO2
figure;
loglog(pd, V_Air, 'b', 'DisplayName', 'Air');
```

```

hold on;
loglog(pd, V_Ar, 'r', 'DisplayName', 'Argon');
loglog(pd, V_CO2, 'g', 'DisplayName', 'CO2');
y_air_pts = interp1(pd, V_Air, x, 'makima');
scatter(x, y_air_pts, 'filled', 'DisplayName', 'Selected Paschen Curve Points');
text(x, y_air_pts, labels, 'VerticalAlignment', 'top', 'HorizontalAlignment',
'left')
hold off;
grid on;
xlabel('pd (torr \cdot cm)');
ylabel('Voltage (V)');
title('Paschen Curves for Different Gases');
text(0.98, 0.02, 'Joshua Davidov', ...
    'Units', 'normalized', ...
    'HorizontalAlignment', 'right', ...
    'VerticalAlignment', 'bottom', ...
    'FontSize', 10);
legend('show');

% Plot Paschen curves for Air only, varying gamma

gamma_vals = 10.^(-5:-1); % [10^-5, 10^-4, 10^-3, 10^-2, 10^-1]

figure;

for i = 1:length(gamma_vals)
    gamma_i = gamma_vals(i);

    denom = log(pd) + log(A_Air) - log(log((1 / gamma_i) + 1));

    V_Air_gamma = B_Air .* pd ./ denom;
    V_Air_gamma(denom <= 0) = NaN;

```

```
loglog(pd, V_Air_gamma, ...
       'LineWidth', 1.5, ...
       'DisplayName', sprintf('\gamma = 10^{\%d}', log10(gamma_i)));

hold on;

end

hold off;
grid on;

xlabel('pd (torr \cdot cm)');
ylabel('Voltage (V)');
title('Paschen Curve for Air at Different Secondary Emission Coefficients');

text(0.98, 0.02, 'Joshua Davidov', ...
     'Units', 'normalized', ...
     'HorizontalAlignment', 'right', ...
     'VerticalAlignment', 'bottom', ...
     'FontSize', 10);

legend('show', 'Location', 'northeast');
```

5. Appendix Typst Code

If you're interested in seeing the code I used to produce this document, here's my template .typ file:

```
#show figure: set block(below: (2.0em))

#let numbered_eq(content) = math.equation(
  block: true,
  numbering: "(1)",
  content,
)

#let date = datetime.today()

#show table.cell.where(y: 0): set text(weight: "bold")

#let conf(
  course_code: [XYZ 123],
  course_title: [Course Title],
  course_color: gradient.linear(..color.map.plasma),
  authors: ("Allen Rakhimov", "Additional Authors"),

  title: [Document Title],
  instructors: ("Professor 1", "Professor 2",),
  month: [datetime.now().month.repr:long],
  year: [2025],
  doc,
) = {
  set page(
    paper: "us-letter",
    margin: (1in),
  )

  set text(
    font: "New Computer Modern",
    size: 12pt
```

```
)

set par(
  leading: 1.30em
)

show link: url => underline(text(fill: rgb(0, 0, 255))[#url])

let date = [#month #year]
let left_head = [#course_code: #title]
let course = [#course_code: #course_title]

// Dynamic right header :)
let right_head = [
  Joshua Davidov \
]

// Course Title
text(course_color, size: 24pt)[
  #box(course)
]

v(1em, weak: true)

for author in authors [
  #if author != authors.at(-1) [
    #author
  ] else [
    #author
  ]
]

// Document Title
place(
```

```
horizon + center,
text(size: 24pt)[#title]
)

place(
bottom + right,
[
  #if instructors.len() > 1 [
    Instructors:
  ] else [
    Instructor:
  ]
  #for instructor in instructors [
    #if instructor != instructors.at(-1) [
      #instructor\
    ] else [
      #instructor
    ]
  ]
]

#date
]
)

pagebreak()
// Create header but only after title page
counter(page).update(1)
set page(
  numbering: "1",
  header: [
    #set text(10pt)
    #left_head
    #h(1fr) Davidov, J. 2026
  ]
)
```

```
#v(0.6em, weak: true)
#line(length: 100%, stroke: 0.5pt)
],
)
doc
}
```

6. References

- [1] A. Fridman, Plasma Chemistry. Cambridge, U.K.: Cambridge Univ. Press, 2008.
- [2] Y. Kemari, C. van de Steen, G. Belijar, L. Laudebat, S. Diaham, Z. Valdez-Nava, and C. Abadie, “A Townsend’s secondary ionization coefficient estimation method for partial discharge inception voltage prediction for insulating polymers,” in *Proc. 2022 IEEE 4th Int. Conf. on Dielectrics (ICD)*, Palermo, Italy, Jul. 2022, pp. 226–229, doi: 10.1109/ICD53806.2022.9863604.